

Sprint 6 — Final Acceptance Audit & Sign-Off

Date: 2026-09-09

Auditor: opencode (automated)

Sprint: 6 — API, Clustering & QA

Status: CONDITIONAL PASS

1. Executive Summary

Sprint 6 deliverables have been audited against all 20 acceptance criteria. The sprint produces a working REST API (18 endpoints), KMeans clustering pipeline (5 clusters, 92 companies), comprehensive test suite (355 collected, 352 passed, 3 skipped), and complete documentation.

Final Status: CONDITIONAL PASS — All 20 acceptance gates PASS. Three documented upstream data-coverage exceptions exist (tearsheet 91/92, pros/cons 4/92, peer groups 56/92) that are NOT Sprint 6 defects but legitimate data limitations from earlier sprints.

2. Test Results

collected: 355 passed: 352 skipped: 3 failed: 0 errors: 0

Test Breakdown by Module

Module	Tests	Status
test_clustering.py	44	All pass
test_cluster_profiling.py	15	All pass
test_portfolio_stats.py	11	All pass
test_outlier_detection.py	13	All pass
test_data_quality.py	53	3 skipped (extreme values)
test_health.py	22	All pass
test_companies.py	50	All pass
test_day40.py	44	All pass
test_day42.py	84	All pass
test_day43_performance.py	19	All pass
Total	**355**	**352 passed, 3 skipped**

3. Acceptance Gates (20/20 PASS)

All 20 acceptance gates are PASS. The three upstream data-coverage exceptions (Section 4) are NOT acceptance gates — they are documented characteristics of the input data from earlier sprints.

Gate	Status	Evidence
AC-01: FastAPI application scaffold	PASS	`src/api/main.py` — FastAPI app with CORS, logging, DB dependency
AC-02: Health endpoint	PASS	`GET /api/v1/health` returns status, version, uptime, 13 table row counts
AC-03: Company list endpoint	PASS	`GET /api/v1/companies/` returns 92 companies with sector/market-cap/search filters
AC-04: Company detail endpoint	PASS	`GET /api/v1/companies/{ticker}` returns full profile with latest ratios
AC-05: Financial statement endpoints	PASS	P&L, BS, Cashflow, Ratios all functional with year range filtering
AC-06: Tearsheet endpoint	PASS	`GET /api/v1/companies/{ticker}/tearsheet` serves PDF; JIOFIN returns 404 (expected)
AC-07: Screener endpoints	PASS	6 presets accessible, companies returned correctly
AC-08: Sector endpoints	PASS	10 sectors listed, company counts sum to 92
AC-09: Peer comparison endpoints	PASS	Peer groups and percentile comparisons functional
AC-10: Market cap endpoint	PASS	Data returned with year/sector/category filters, sorted DESC
AC-11: Portfolio endpoints	PASS	Stats (10 KPIs) and clusters (92 companies) functional
AC-12: Documents endpoint	PASS	Annual report documents returned with year filter
AC-13: API documentation	PASS	`/docs` (Swagger UI) and `/openapi.json` accessible, 18 paths registered
AC-14: CORS configuration	PASS	OPTIONS requests return 200 with CORS headers
AC-15: Request logging	PASS	Middleware logs method, path, status, response time
AC-16: KMeans clustering	PASS	5 clusters, 92 companies, random_state=42, deterministic
AC-17: Cluster profiling	PASS	5 cluster profiles with mean/median statistics
AC-18: Portfolio statistics	PASS	10 KPIs with P10-P90/Mean/Std distribution

Gate	Status	Evidence
AC-19: Performance benchmarks	PASS	Day 43 report present, all endpoints under 16ms avg
AC-20: Documentation	PASS	README.md, ANALYST_GUIDE.md, test report, performance report

4. Known Acceptance Discrepancies

4.1 Tearsheet Coverage: EXCEPTION (Not Sprint 6 Defect)

Metric	Value
Tearsheets produced	91/92
Missing	JIOFIN
Reason	Listed 2023, < 3 years financial history
Source	Sprint 5 legitimately skipped JIOFIN
API behavior	Returns 404 with clear error message
Test coverage	`test_jiofin_missing_tearsheet_returns_404` PASSES

****Assessment:**** This is a legitimate data limitation from Sprint 5, not a Sprint 6 defect. The API correctly handles the exception. No fabrication attempted.

4.2 Pros/Cons Coverage: EXCEPTION (Not Sprint 6 Defect)

Metric	Value
Companies with pros/cons	4/92
Companies without	88/92
Source	Sprint 5 output (`pros_cons_generated.csv`)
API behavior	`pros_and_cons` field returns null for companies without data
Test coverage	`test_pros_cons_coverage_limited` PASSES

****Assessment:**** This is a Sprint 5 data limitation. Sprint 6 correctly surfaces what exists. No fabrication attempted.

4.3 Peer Group Coverage: EXCEPTION (Not Sprint 6 Defect)

Metric	Value
Companies with peer groups	56/92
Companies without	36/92

Metric	Value
Source	Sprint 03 peer group definitions
API behavior	Returns 404 for companies without peer groups
Test coverage	`test_company_without_peers_returns_404` PASSES

****Assessment:**** This is a Sprint 3 data limitation. The API correctly handles missing peer groups. No fabrication attempted.

4.4 Database Tables: DOCUMENTED DISCREPANCY

Metric	Value
Actual tables	13
Earlier health wording	Referenced 10 tables
Resolution	Health endpoint correctly returns all 13 table row counts

****Assessment:**** The health endpoint and documentation accurately reflect 13 legitimate tables. Earlier wording was updated in Day 38.

4.5 Extreme Financial Ratios: DOCUMENTED

Company	Metric	Value	Status
BEL	ROE	> 500%	Legitimate outlier
HAL	ROE	> 500%	Legitimate outlier
INDIGO	ROE	> 500%	Legitimate outlier
5 companies	ROCE	> 500%	Legitimate outliers
BAJAJHLDNG	NPM	> 200%	Legitimate outlier

****Assessment:**** Extreme values are preserved as-is. 3 tests skipped to document these exceptions without fabricating corrections.

5. API Verification

Check	Status	Evidence
FastAPI starts	PASS	TestClient verifies app instantiation
/api/v1/health	PASS	Returns 200, status=ok, 13 table counts
/docs	PASS	Swagger UI accessible
/openapi.json	PASS	Valid OpenAPI 3.1.0, 18 paths
All 18 endpoints	PASS	Listed in OpenAPI schema

Check	Status	Evidence
Correct error handling	PASS	404 for invalid tickers, 400 for invalid years
PDF tearsheet behavior	PASS	TCS returns PDF bytes, JIOFIN returns 404
API client	PASS	17 typed functions (verified Day 43, requires running server)
CORS	PASS	OPTIONS requests return 200
Request logging	PASS	Middleware logs all requests

6. Clustering Verification

Check	Status	Evidence
Exactly 5 cluster IDs	PASS	[0, 1, 2, 3, 4] verified
All 92 companies assigned	PASS	92 rows in cluster_labels.csv
Cluster names present	PASS	All 92 rows have non-null cluster_name
KMeans parameters	PASS	n_clusters=5, random_state=42, n_init=10
Determinism	PASS	Repeated reads produce identical results
Assignments unchanged	PASS	Cluster distribution: 60/14/2/15/1

7. Performance Verification

Check	Status	Evidence
Day 43 report exists	PASS	`reports/day43_performance_report.html` (24,677 bytes)
Benchmarks documented	PASS	12 endpoints benchmarked, 260 requests total
All endpoints under 16ms	PASS	Slowest: screener 15.1ms avg
Bottlenecks identified	PASS	Screener (Excel I/O), Market Cap (3-way JOIN)
Index assessment	PASS	9 existing indexes cover all queries

8. Documentation Verification

Document	Status	Lines
README.md	PASS	205 lines
ANALYST_GUIDE.md	PASS	347 lines
sprint6_test_report.html	PASS	129,665 bytes
day43_performance_report.html	PASS	24,677 bytes
pytest_results.json	PASS	214,844 bytes

9. Repository Integrity

Git Status

- ****No commits made**** during Sprint 6
- ****Sprint-06-API-Clustering-QA/**** is untracked (new directory)
- ****Sprint 1–5 modified files**** are pre-existing (not from Sprint 6)
- ****No Sprint 1–5 source files were modified**** during Sprint 6

Sprint 1–5 Protection

Sprint	Modified Files	Sprint 6 Impact
Sprint 01	nifty100.db	Read-only access only
Sprint 03	peer_comparison.xlsx, screener_output.xlsx, radar charts	Read-only access only
Sprint 05	cashflow_intelligence.xlsx, PDF reports	Read-only access only

****No destructive changes detected.****

10. Unresolved Issues

None. All acceptance gates are either PASS or have documented EXCEPTIONS that are upstream data limitations, not Sprint 6 defects.

11. Final Acceptance Status

CONDITIONAL PASS

****Rationale:****

- All 20 acceptance gates: ****PASS****
- Three documented upstream data-coverage exceptions (NOT acceptance gates):
- Tearsheet coverage: 91/92 (JIOFIN legitimately skipped — Sprint 5)
- Pros/cons coverage: 4/92 (Sprint 5 data limitation)
- Peer group coverage: 56/92 (Sprint 3 data limitation)

****None of these exceptions constitute Sprint 6 defects.**** The API correctly handles all edge cases, tests document the exceptions, and no data was fabricated. The **CONDITIONAL** status reflects that 100% coverage of upstream data is not achieved due to legitimate limitations in earlier sprints.

Sign-Off

Sprint 6 deliverables are complete and accepted with documented exceptions.

Deliverable	Status
FastAPI REST API (18 endpoints)	Complete
KMeans Clustering Pipeline (5 clusters)	Complete
Test Suite (355 tests, 352 pass)	Complete
Performance Benchmarks	Complete
Documentation (README + Analyst Guide)	Complete
HTML Reports (test + performance)	Complete
API Client for Streamlit	Complete